

Additional Issues/Suggestions/Comments/Questions

General suggestions for Section 4

1. **Shorten the examples where possible.** Section 4 contains many code blocks. Retain code that teaches the public API: model construction, fitting, diagnostics, prediction, synthesis, and standard plotting methods. Consider moving detailed graphics settings, manual legend construction, forecast-matrix assembly, and table-construction code to the replication materials or an appendix.

2. **Runtime comparisons and implementation details.** Give consistent computational details for runtime tables (you can do this once at the beginning and say it is used elsewhere) and emphasize that runtimes are machine- and implementation-dependent. The predictive, diagnostic, and recovery metrics should carry the substantive comparison.

Backend routing, thread counts, cache management, and output paths are important for reproducibility but can interrupt the example narrative. Keep the essential computational information in the article and place detailed operational information in the replication documentation or appendix.

3. **Question and figure list: MAP versus posterior mean.** The installed plotting implementation computes the fitted quantile, component, and state lines using `rowMeans()` across the stored posterior or variational draws. Although the internal returned fields are named `map.quant` and `map.comp`, these calculations produce posterior means rather than posterior modes. Please verify this against the package version used for the manuscript and revise every affected caption and nearby sentence from “MAP estimates” to “posterior mean estimates.” The affected figures are:

- the Lake Huron figure `ex1quants.png`, specifically the fitted dynamic quantiles in its top-left panel;
- the Sunspots figure `ex2quant.png`, specifically the fitted DQLM and `exDQLM` quantiles in its bottom-left panel;
- the Big Tree figure `ex3quantcomps.png`, including the fitted quantiles, seasonal components, and covariate contributions; and
- the Big Tree figure `ex3zetapsi.png`, including the displayed ζ_t , $\psi_{\text{NOI},t}$, and $\psi_{\text{AMO},t}$ state summaries.

Do not change terminology mechanically in the diagnostic and forecast figures. The Sunspots diagnostic figure `ex2checks.png` uses MAP plug-in standardized forecast errors as defined in Section 2.2, so “MAP plug-in” is appropriate there. The Big Tree forecast figure `ex3forecast.png` uses filtering means for the historical portion and forecast means for the future portion, so “posterior mean forecast quantiles” is appropriate. Please also consider whether the misleading

internal field names should be documented or renamed separately, taking backward compatibility into account.

Section 4.1: Lake Huron

1. **Clarify which median model is used.** The object `M50` is initially assigned an unrestricted exDQLM fit and is later overwritten by a fit with `fix.gamma = TRUE`. Therefore, the final median plot, forecast, and synthesis use the fixed- γ DQLM fit, while the tail fits use exDQLMs. Either rename the second object (for example, `M50.dqlm`) or state explicitly why the fit is replaced and which version is used in subsequent results.
2. **Remove or justify the separate `M50.trace` fit.** The unrestricted median model is fitted twice: `M50` uses 2000 burn-in iterations, whereas `M50.trace` uses 7000 burn-in iterations and a separate seed solely for the MCMC diagnostic figure and the posterior summaries of σ and γ . The article says that the longer burn-in is used “for the diagnostic display,” but it does not explain why it is necessary or show that the original `M50` chain has a convergence problem. In addition, the complete replication workflow appears to fit `M50.trace` and the fixed- γ median model, but not the original unrestricted 2000-burn-in `M50` fit.

The preferred simplification is to remove the original unrestricted `M50` fit, use `M50.trace` as the single unrestricted median fit for assessing γ , and name the subsequent fixed- γ model `M50.dqlm`. Alternatively, if the original `M50` chain has satisfactory convergence, use that chain for the trace and density plots and remove `M50.trace`. If the 7000-iteration burn-in is genuinely required, justify it using an appropriate convergence assessment rather than describing it as necessary only for the display.
3. **Comment out the output printed after the M5 fit.** Keep the displayed call that creates `M5`, but comment out or remove the console and progress output that follows it. The iteration updates and backend messages are not needed to explain the fitting workflow and take up space in the article. They can remain available in the complete replication log.
4. **Question about MCMC initialization.** If `m_gam`, `s_gam`, and `df_gam` in `PriorGamma` specify the prior distribution for γ and do not determine its MCMC starting value, how is the chain actually initialized in these Lake Huron fits? Please verify whether `exdqlmMCMC()` uses the LDVB fit by default to initialize γ , whether it uses `gam.init` when that argument is supplied, and what happens when neither an explicit initial value nor an LDVB warm start is used. The article should clearly distinguish the prior location parameter from the value used to initialize the chain.
5. **Question about quantile crossing and synthesis.** In the Lake Huron analysis, did the separately fitted quantiles at $p_0 = 0.05, 0.50, \text{ and } 0.95$ cross during either the observed period or the forecast period? The article states that isotonic adjustment and monotone rearrangement are available, but the displayed calls to `quantileSynthesis()` do not include an argument selecting a crossing correction. Also state which procedure, if any, was used to produce `syn.obs`, `syn.fore`, and Figure 1. If no crossing occurred, say explicitly that no correction was required. If a correction was applied, identify the method and whether it changed the reported synthesis materially.
6. **Question about mixing model specifications in the synthesis.** The calls to `quantileSynthesis()` combine `M5` and `M95`, which are exDQLM fits with γ estimated, with `M50`, which

has been overwritten by the fixed- γ DQLM median fit. Is the synthesis method intended to combine posterior predictive draws from quantile models with different likelihood specifications in this way? Please verify that the method does not require all input models to share the same treatment of γ and explain why this mixed exDQLM-tail/DQLM-median specification is appropriate. If it is appropriate, state explicitly in Section 4.1 that the synthesis combines the two exDQLM tail fits with the simpler fixed- γ DQLM median fit. This synthesis is not explained before anywhere right?

Section 4.2: Sunspots

1. **Question about agreement between CRPS and KL.** The article states that CRPS and KL both select the seasonal discount factor 0.85, but the displayed output prints only the row selected by CRPS. How can the reader verify that the KL minimum occurs in the same row? Please either display a compact table containing every candidate seasonal discount factor and its CRPS and KL values, print the selections from both criteria, or remove the statement that the two criteria agree. Also confirm that the values shown in the article are generated by the same recorded run used for the surrounding Sunspots results.
2. **Question about `sig.init = 2`.** Why is `sig.init = 2` supplied to every LDVB fit in the discount-factor comparison when it is not supplied to the principal M2 fit shown earlier? Please verify how the principal fit initializes σ , whether the candidate fits and the principal fit should use the same initialization, and whether changing `sig.init` affects LDVB convergence, the resulting CRPS and KL values, or the selected seasonal discount factor. If the value 2 is required for numerical stability, explain and justify it in the article. If it is not required, remove it so that the candidate-model comparison matches the main exDQLM specification more closely.
3. **Question about random-number seeds.** What random-number seeds were used to produce the reported Sunspots LDVB fits, the posterior draws from the variational approximations, the candidate discount-factor fits, and the matched MCMC fits? The displayed code does not call `set.seed()`, even though the posterior sampling and MCMC results are stochastic. Please identify the seeds used by the replication scripts and add explicit, appropriately placed `set.seed()` calls to the reader-facing code. Confirm that the seeds in the manuscript, `examples.R`, and the full replication workflow correspond to the results in the figures, printed output, and benchmark table.
4. **Question about inconsistent Sunspots results.** Why do the values printed by `print(diagM1M2)` differ from the LDVB values reported in Table 2 when both appear to describe the same DQLM and exDQLM fits with `n.samp = 3000`? The printed output reports DQLM and exDQLM runtimes of 24.991 and 128.725 seconds, respectively, whereas Table 2 reports 52.65 and 251.81 seconds. The diagnostic values also differ: the printed output gives CRPS values 8.5460 and 6.2551 and PPLC values 3853.4604 and 2331.6722, while Table 2 gives CRPS values 8.523 and 6.250 and PPLC values 3852.3 and 2326.2. Please identify the exact run, seed, machine, package version, initialization, and backend profile that generated each set of numbers. Ideally, regenerate the printed output and Table 2 from the same recorded fitted objects. If two different runs or environments must remain, label them explicitly and explain why both sets are shown so readers do not interpret the discrepancy as a reproducibility error.

Section 4.3: Big Tree water flow

1. **Question about LDVB tolerance and posterior sample size.** Why do the Big Tree LDVB fits use `tol = 0.05` and `n.samp = 1000`? Briefly explain precisely what the tolerance controls in the implementation and verify that 0.05 provides adequate convergence for `M0`, `MREG`, `MTF`, and every candidate value of λ . Also check whether the fitted quantiles, component summaries, PPLC values, selected λ , and held-out forecast scores remain stable under a tighter tolerance and a larger number of posterior draws. If these settings were chosen primarily to reduce runtime for the example, state that explicitly and report the sensitivity check in the replication materials.
2. **Question about the held-out forecast results.** How should the held-out results in Table 4 be interpreted? The direct-regression model `MREG` has the smallest check loss and CRPS, the no-covariate model `M0` is second, and the transfer-function model `MTF` has the largest value for both metrics. Please add a direct discussion of this ranking rather than ending with the in-sample diagnostic comparison. In particular, explain that selecting λ using training-period PPLC did not make `MTF` the best held-out forecasting model in this example, and verify whether the ranking is stable to the LDVB tolerance, posterior sample size, and Monte Carlo seed.
3. **Question about the purpose of the transfer-function comparison.** Is the purpose of this example to demonstrate the package’s transfer-function construction and forecasting workflow, or to claim that transfer functions improve prediction for these data? The held-out results do not support a claim of predictive superiority for `MTF`. Please state explicitly that `MREG` gives the best held-out scores here, while `MTF` illustrates how persistent lagged covariate effects can be specified, inspected, and forecast with the package. The conclusion should distinguish demonstration of a software capability from evidence that the more elaborate model is preferred for this dataset.

Section 4.4: Static exAL regression

1. **Question about the shrinkage-prior controls.** What roles do `a_zeta = 2` and `b_zeta = 9` play in the Nishimura–Suchard regularized horseshoe implementation when `zeta2_fixed = 9` is also supplied? The prose explains `tau0 = 0.15`, the fixed slab $\zeta^2 = 9$, and the unshrunk intercept, but does not explain `a_zeta` or `b_zeta`. Please verify from the implementation whether these two hyperparameters remain active when the slab variance is fixed. If they are active, explain the distribution they parameterize and why the values 2 and 9 are used. If they are ignored under `zeta2_fixed = 9`, remove them from the displayed `rhs.ctrl` list so that readers are not led to believe that they affect the reported fits.
2. **Question about `n_samp_xi = 160`.** What quantity is sampled or approximated through `n_samp_xi` in `exalStaticLDVB()`, and why is its value set to 160? Is the equality with the training sample size intentional or coincidental? Please explain how this Monte Carlo setting affects the LDVB objective or updates and verify that the coefficient summaries, diagnostic metrics, and holdout quantile RMSE are stable under a larger value. If 160 is a package default or a computational compromise, state that clearly.
3. **Question about LDVB convergence.** Did all three static LDVB fits actually satisfy the convergence criterion, and how many iterations did each fit require? The text says that the $p_0 = 0.05$ fit converges more slowly and therefore receives `max_iter = 420` rather than 260, but a larger iteration budget does not by itself demonstrate convergence. Please report or reference the returned convergence indicators and iteration counts, explain the criterion associated with

`tol = 1e-4`, and confirm that the results are stable when the maximum iteration limits are increased.

4. **Question about MCMC adequacy.** What evidence shows that 2000 burn-in iterations and 3000 retained draws are adequate for each static MCMC fit? Because the MCMC chains are initialized from the corresponding LDVB fits and are treated as the posterior-simulation comparison, please examine effective sample sizes, Monte Carlo standard errors, trace behavior, and, if available, multiple-chain diagnostics. Also verify that the retained posterior summaries are not materially sensitive to the LDVB warm start. The article can refer readers to diagnostics in the replication materials, but it should not call MCMC the reference approach without evidence that the chains mixed adequately.
5. **Question about runtime.** Again, under what computational environment were the LDVB and MCMC runtimes in the static benchmark table measured? Please identify the processor, operating system, R version, `exdqlm` version, computational backend and mode, and number of threads, and confirm whether the reported times include only model fitting or also initialization, diagnostics, and posterior-summary construction. Use the same provenance convention as the Sunspots benchmark, or refer both tables to one common benchmark-environment statement and output file. Also verify that every runtime in the table was generated under the same profile so that the within-quantile LDVB–MCMC comparisons are meaningful.